

UNDERSTANDING THE HETEROGENEITY OF EARNINGS LOSSES AFTER JOB DISPLACEMENT: A MACHINE-LEARNING APPROACH*

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Abstract

Using generalized random forests in a difference-in-differences framework, we analyze heterogeneity in earnings losses following job displacement. We find substantial variation in short-term losses ranging 20-70% of pre-displacement earnings. While all workers face long-term losses, employment rather than wage changes primarily drives heterogeneity. The most important predictor for employment losses is worker’s age, while the firm wage premium is the most important for wage losses. We predict similar loss patterns for the population excluded by the typical sample restrictions in the literature. We further show how to create simple decision rules to target active labor market programs to high-loss individuals.

Keywords: Job displacement, Earnings losses, Causal machine learning

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I. INTRODUCTION

A sizable literature documents that, on average, workers displaced during a mass layoff experience significant losses in annual income lasting over 15 to 20 years.¹ Many labor market policies are designed to avoid job losses in general. Prominent examples include firm bailouts, employment protection, and employment subsidies such as short time work schemes, which are often applied in a non-discriminatory fashion. Critics argue that not all jobs are worth saving, and such programs should be, if anything, targeted. To design effective policy responses, we not only need to identify the consequences of job loss for the average worker, but also understand how earnings losses differ across individuals, and which pre-displacement characteristics lead to higher losses. While many papers show that earnings losses differ by certain worker or firm characteristics, this paper systematically documents the heterogeneity of short- and long-term cost of job loss across the population and identifies the most important predictors for losses.

To this end, we draw on recent advancements in machine learning to document the distribution of earnings, employment, and wage losses conditional on workers' and firms' characteristics. We implement a generalized random forest by [Athey, Tibshirani, and Wager \(2019\)](#) to a difference-in-difference (hereafter DiD) setup. Furthermore, we study how post-displacement employment histories differ across individuals with high and low predicted losses and discuss which pre-displacement characteristics are the most important predictors for the cost of job loss. We then show how our results could be used to design simple decision rules for targeting displaced workers predicted to face especially severe employment and wage losses.

Our approach estimates the conditional average treatment effect (CATE) of job loss nonparametrically as a function of observables. This is achieved by exploring possible data splits and choosing those ones for which between-group differences in losses are the highest. By recursively splitting the dataset into smaller and smaller subsamples, the algorithm builds a tree which detects heterogeneity in the difference-in-difference estimator of the cost of job loss. Thus, we need to assume that the outcome variables for the treatment group absent of job loss would have evolved in parallel to the control group for every subsample. We verify that this assumption is plausible with several diagnostic checks. To avoid detection of

¹Jacobson et al. (1993), Neal (1995), Couch and Placzek (2010), Davis and Von Wachter (2011), Farber (2011), Farber (2017), among many others. Furthermore, job displacements have been shown to have detrimental effects on health (Schaller and Stevens, 2015), longevity (Sullivan and Von Wachter, 2009), and on children of displaced workers (Lindo, 2011; Rege et al., 2011).

artificial heterogeneity, instead of growing a single tree, we build a random forest consisting of many tree-based models. Every tree is trained using a random subset of observations and variables and then their estimates are combined.² Another appealing property of random forests is that we are able to quantify standard errors arising from two sources, the machine-learning procedure and the estimation procedure.

We use the universe of Austrian social security records over three decades and apply the typical identification strategy for job displacements by focusing on mass-layoffs. Displaced Austrian workers suffer on average losses of similar magnitude compared to those displaced in Germany (Schmieder et al., 2023) and the United States (Davis and Von Wachter, 2011).

Our algorithm documents losses conditional on workers’ and job characteristics. For this purpose, we construct 58 partitioning variables to study how earnings losses vary with pre-displacement worker, firm, time, and local labor market characteristics and to identify the most important predictors for those losses.

We build separate forests for short- (one year after job loss) and long-term (10 years after job loss) losses in relative earnings, employment, and log-wages. We use these forests to estimate a CATE of a job loss for every individual in our sample. Based on observable characteristics, almost no worker is predicted to have losses lower than 20% in the year after the job loss, while the worst affected workers face CATEs of more than 70%. The 25th percentile of CATEs is 42%, while the 75th is 29%. This heterogeneity is not driven by noisy estimates of the CATEs. At the 95% confidence level, we show that 67% of individual CATEs are different compared to the median level of 35%. The distribution of long-run CATEs shifts towards lower losses, and its dispersion contracts markedly. 25% of workers face predictable losses of more than 19% in the long-run and the interquartile range decreases to 6 percentage points (pp). In our sample, 99% of all workers are predicted to experience losses in the long-run. Thus, in line with recent evidence from the United States (Rose and Shem-Tov, 2023), job loss is costly even for low wage workers with disadvantaged worker and firm characteristics. Although there is some convergence of losses across workers, the correlation of short and long-term CATEs is fairly high with 0.4.

Earnings losses are a result of both forgone earnings due to non-employment and declines in wages after reemployment. Overall the predicted earnings losses are more strongly correlated with employment than with wage losses, indicating that the dispersion in employment

²This statement is for expositional reasons. Strictly speaking, for an individual characterized by observables $\mathbf{z}$, the random forest provides weights $\alpha_i(\mathbf{z})$ measuring similarity of all other observations indexed by i . Those weights are used in weighted linear regression to estimate the causal effect at $\mathbf{z}$. The whole algorithm is presented in Subsection IV.A in greater detail.

losses is a somewhat more important driver of the heterogeneity in overall losses. The 25th and 75th percentile of predictable short-run employment losses are 125 and 85 days. In the long-run, only 25% of CATEs are more than 37 lost employment days per year, the interquartile-range narrows significantly to 16 days. Predictable wage losses in the short run range from 35% to even wage gains. Interestingly, the distribution of predictable wage losses in the short and long-run are very similar, suggesting that wage losses are very persistent. This is confirmed by the high correlation of close to 0.8 for short and long-term CATEs in wage losses.

How do labor market histories after job loss differ across individuals with high and low CATEs in earnings losses? Those individuals identified to face the highest losses have the lowest propensity to remain in their original industry, and move to more unstable firms, and subsequently have more job changes. They also suffer the largest declines in employer wage premia. Interestingly, we find that across the earnings-loss distribution, declines in employer wage premia account for a roughly constant 60-70% of wage losses.

What pre-displacement worker and job characteristics are the most important predictors for the severity of losses? We compute how predicted earnings, employment, and wage losses change from the lowest to the highest realization of the covariates, holding every other observable variable constant at the median. The variable that predicts the highest change in short-term losses is age, the oldest workers face losses of 50%, whereas the youngest workers with the same worker and firm characteristics face losses of 30%. Firm size and the firm wage premia are the second and third most important predictors of short-term earnings losses. For long-run earnings losses, the firm’s labor-market share and firm separation rate prior to the mass-layoff event are the first and second most important variables, followed by workers’ age. *Ceteris paribus*, workers from firms with the largest labor market share or most stable employers face 7-8pp higher earnings losses 10 years after job loss, highlighting the special role of job losses at regional flag-ship firms (Gathmann et al., 2020) and particularly stable jobs (Jarosch, 2023). As employment losses are strongly correlated with overall earnings losses, the variable importance is similar for employment losses.

Fackler, Mueller, and Stegmaier (2021), Schmieder, Von Wachter, and Heining (2023), and Bertheau, Acabbi, Barceló, Gulyas, Lombardi, and Saggio (2023) find that losses in firm wage premia account for a large fraction of wage losses, whereas Lachowska, Mas, and Woodbury (2020) find a limited role of firm wage premia. We shed light on this discussion from a new angle: Is the pre-displacement firm wage premium an important predictor for subsequent earnings, employment, and wage losses? Although the firm wage premium is

not among the most important predictors for employment losses, it is the most important variable for short and long-run wage losses. Workers displaced from firms with wage premia in the highest decile are predicted to face short-term wage losses of 18pp and long-run losses of 9pp more than those displaced from the lowest decile firms. Thus, for each log point higher pre-displacement firm fixed effects, long-run wage losses increase by roughly 0.15 log points.

Interestingly, many variables that the prior literature has shown to matter for earnings losses such as business cycles (Davis and Von Wachter, 2011; Schmieder et al., 2023), manufacturing (Helm et al., 2023), job tenure (Jacobson et al., 1993; Neal, 1995) do not show up as important predictors of losses in our setting.

In principal our results could be used to target high-loss individuals with active labor-market programs. A disadvantage is that we use variables that are constructed using statistical measures (*e.g.*, the fixed effects) which may not be readily understandable by the general public and conveniently observable by policy makers. To address this, we use comprehensible variables to build a transparent and clearly communicable decision tree to target high employment- and wage-loss individuals. They correctly identify approximately 75-77% of individuals with above-median predicted losses.

We compare our generalized random forest approach with alternatives. LASSO-penalized DiD regression shows greater variation in treatment effects but identifies fewer individuals with statistically significant differences from the median (21% compared to our forest’s 67%). Quantile-based approaches struggle with individual-level effects due to poor correlation between pre- and post-displacement earnings ranks and a simple comparison of control and treatment groups does not give a sense of statistical accuracy and does not identify the most important predictors. The comparison highlights that our forest-based approach is the best for analyzing predictable individual losses.

In addition to previously mentioned papers, we also contribute to the small but growing number of papers using machine learning to study heterogeneous treatment effects in labor economics.³ Gregory, Menzio, and Wiczer (2025) and Castro, Lange, and Poschke (2025) use a k -means algorithm to classify workers into three clusters based on labor market histories and document the heterogeneity in earnings losses for these three groups. We document the whole distribution of losses and identify pre-displacement worker and job characteristics that are most important predictors of earnings losses. In a recent working paper, Athey et al. (2023) use a similar machine-learning algorithm to ours to study heterogeneity in earnings

³*E.g.*, Davis and Heller (2020); Knaus et al. (2022).

losses in Sweden. They also find that worker’s age is the most important predictor for short-term earnings losses. We go beyond their study and examine both short and long-term earnings losses as well as log-wage losses. We show that the most important predictors for long-term losses shift from worker’s age towards firm characteristics. Firm wage premium, not present in their study, is by far the most important predictor for wage losses. In addition, we leverage our forest to address a long-standing concern in the literature: Do the strict sample restrictions for a clean identification of the displacement event select individuals that are bound to experience large earnings losses? We document that the distributions of predicted long-term earnings losses for the general population and for displaced workers in our sample are surprisingly similar.

The rest of the paper is organized as follows. The next section describes the empirical setting in Austria, as well as the sample selection. Section III presents the average cost of job displacement. Section IV describes our machine-learning algorithm, and Section V documents the heterogeneous scarring effects of job displacement and discusses their most important predictors. Section VI demonstrates how our results can be translated into practical targeting strategies using transparent decision rules, while Section VII juxtaposes our generalized random forest approach with alternative methods for analyzing treatment effect heterogeneity. The last section concludes.

II. EMPIRICAL SETTING

We use Austrian social security records from 1984 through 2020, which comprise day-to-day information on all jobs and unemployment spells covered by social security ([Arbeitsmarktdatenbank, 2021](#)). It contains all private sector jobs and a large fraction of public sector employment, but excludes self-employed and public servants who are not covered by social security. It contains yearly earnings for each worker-establishment pair, and basic socio-demographic information such as age, gender, a flag for a blue collar occupation, and citizenship.⁴ Each establishment (we use firm and establishment exchangeably from here on) has a unique identifier, we have information on its geographic location, age, and 4-digit industry.

⁴We deflate all earnings to 2017 level using the CPI index provided by the Austrian Statistical Agency.

A. Definition of Job Displacement and Mass Layoff

To ensure comparability with the literature on displaced workers, we follow the typically applied definitions and sample restrictions as much as possible. Workers are considered displaced if they separated from their primary employer that experienced a mass layoff in the given year. We define a mass layoff event at the firm level in year t if the size of the firm declined by more than 30% during year t . To avoid selecting volatile firms, we exclude firms that grew by more than 30% in either $t - 1$, or $t - 2$, as well as firms that are larger 3 years after the event than before. To have a meaningful measure of firm growth, we only consider establishments with at least 30 employees. To avoid misspecifying mergers, outsourcing, or firm restructures as mass-layoffs, we compute a worker cross flow matrix for all firms in each year. We exclude all firms where more than 20% of their workforce ends up working for the same employer in $t + 1$, excluding firms with large worker flows to other firms.⁵ We also disregard mass layoff events from the public administration (Nace, Level 1, Code O).

B. Sample Construction

We depart from most of the earnings loss literature and do *not* restrict our sample to males. Specifically, we are interested in how earnings losses vary across different individuals based on their characteristics, including gender. We select everybody who is employed on the reference day of January 1st each year between 1989 and 2010. We follow the literature and restrict our sample to workers aged 24-50, employed at a firm larger than 30 employees, and with job tenure longer than 3 years on the reference day.⁶ We further restrict our sample to individuals who have annual earnings of at least €7,500 for two consecutive years to exclude workers working only a few hours per week in so called mini-jobs. This results in 11,804,786 person-year observations. A common critique is that these sample restrictions are selecting workers that are bound to face high earnings losses. We will use our machine-learning algorithm to make out-of-sample predictions on earnings losses for the general population to study whether these restrictions indeed select worker and job characteristics that are associated with higher losses.

Out of these remaining observations, we define a person to be displaced in t if she separates from a firm experiencing a mass layoff, and she is not reemployed at the same firm

⁵For an in-depth discussion, see [Hethey-Maier and Schmieder \(2013\)](#).

⁶We only consider workers aged below 50 because of the low legal and effective retirement age in Austria. Since we follow workers for 10 years after their job loss, and workers retire on average slightly before the age of 60, we have chosen age 50 to minimize confounding earnings losses with early retirement of both the control and treatment group.

at any point in the next 10 years. If a worker suffers multiple mass layoffs, we only consider the first one. We identify 50,207 displaced worker events between 1989 and 2010.

Some workers disappear from our dataset over time. This happens because workers might not find employment subject to social security insurance anymore and drop out of the labor force, if workers move into self-employment, or move abroad. For those years without any social security spell, we impute zero earnings.

Note that we do not restrict the control group to have stayed at their employer after t . The potential comparison group consists of non-displaced workers subject to the same sample restrictions. This includes workers employed at firms without any mass layoff event during year t or workers in mass layoff firms who did not separate.

Non-displaced workers differ in many characteristics from the displaced workers, as can be seen in Table 1. Following many papers in the literature, e.g. Schmieder et al. (2023); Bertheau et al. (2023); Halla et al. (2020), we use propensity score matching in order to obtain a control group that is as similar as possible to displaced workers. In each year, for all workers satisfying the sample restrictions, we estimate the propensity to experience a displacement event as a function of the following worker and firm characteristics: worker’s log-wage in year $t - 1$ to $t - 3$, tenure, age, establishment size in year $t - 1$, a dummy for working in the production sector, a female indicator, a dummy for having a full year of non-employment or no social security spells in the last 5 years, and the number of non-employment days in the last 5 years.⁷ For each displaced worker in a given year, we select the non-displaced worker with the nearest propensity score without replacement. Table 1 shows that our matched control group is very similar to displaced workers in observable characteristics. The two groups are virtually indistinguishable in terms of pre-displacement evolution of earnings, days employed, and log-wages, as can be seen in Figure ?? in the appendix, which plots raw averages across these groups over time.

C. Outcome Variables

We study three outcome variables. First, we estimate how yearly earnings change relative to the average earnings in the two years before job displacement. Second, we measure employment losses as the change in days employed covered by social security. Third, we study how workers’ log daily wage at their dominant employer evolves after job loss conditional on re-employment.⁸ The details for the estimation are found in Appendix ??.

⁷We also experimented with different sets of matching variables, all of which lead to similar results.

⁸We compute daily wages by dividing yearly income from the employer with the highest earnings in a given year and divide it by the number of employment days at this employer.

III. THE AVERAGE COST OF JOB DISPLACEMENT

We start by discussing the estimation strategy for the homogeneous treatment case, which will be extended to heterogeneous treatment effects in the next section. The average cost of job loss for a given post-displacement period h can be compactly estimated using the following equation:

$$y_{it} = \tau_h D_i \times \mathbb{1}(t = t^* + h) + \nu_h \mathbb{1}(t = t^* + h) + \theta D_i + \gamma + \sum_{j=-5}^{-2} \nu_j \mathbb{1}(t = t^* + j) + \sum_{j=-5}^{-2} \delta_j \mathbb{1}(t = t^* + j) \times D_i + \epsilon_{it}, \quad (1)$$

where D_i is an indicator equal to one for a displaced person, t^* is the displacement year and h is the horizon of interest. θ absorbs initial differences in labor market outcomes between the control and treatment group. ν_j and δ_j measure how y_{it} evolves for the control and the treatment group in the years leading up to the event, whereas τ_h measures the cost of job loss h years after job displacement, relative to $t - 1$. We will focus on short-term ($h = 1$) and long-term losses ($h = 10$), and drop the remaining post-treatment horizons. The regression does not feature a year fixed effect, nor a control for age as is typical in the literature, because our machine-learning algorithm will flexibly estimate earnings losses by displacement year and by the age of the workers. Table 2 shows the estimates from this regression for different specifications. For each outcome variable the table reports the cost of job loss using Equation (1) with $h = 1$ without any controls and with worker fixed effects. Table 2 presents regression estimates showing that job displacement leads to significant negative effects across all three outcomes—reducing earnings by 36% relative to pre-displacement levels, employment by approximately 106 days, and log wages by 0.14 — with results reported both without and with worker fixed effects.

Because of the computational burden of the machine learning algorithm, we will not be able to include worker fixed effects as controls as is often done in the literature. But the results from Table 2 show that because our matching procedure selected very similar workers as a control group, adding worker fixed effects does not significantly change the estimated costs of job displacement.

IV. HETEROGENEITY IN THE COST OF JOB LOSS – MACHINE LEARNING APPROACH

The average scarring effects may be a far cry from the experience of most displaced workers. To identify the heterogeneity in earnings losses and its most important predictors, we employ a machine learning procedure built on the methodology of generalized random forests, recently developed by [Athey et al. \(2019\)](#). Conceptually, our approach consists in estimating a version of Equation (1) with heterogeneous scarring effects:

$$y_{it} = \tau_h(\mathbf{z}_i)D_i \times \mathbb{1}(t = t^* + h) + \nu_h(\mathbf{z}_i)\mathbb{1}(t = t^* + h) + \theta(\mathbf{z}_i)D_i + \gamma(\mathbf{z}_i) + \sum_{j=-5}^{-2} \nu_j(\mathbf{z}_i)\mathbb{1}(t = t^* + j) + \sum_{j=-5}^{-2} \delta_j(\mathbf{z}_i)\mathbb{1}(t = t^* + j) \times D_i + \epsilon_{it}, \quad (2)$$

where $\mathbf{z}_i$ are the values of observable variables (henceforth called *partitioning variables*) for individual i . $\tau_h(\mathbf{z}_i)$ is the function that describes how the cost of job loss changes with individual worker and job characteristics $\mathbf{z}_i$. The functional specification of $\tau_h(\mathbf{z}_i)$ is assumed to be unknown. To estimate $\tau_h(\mathbf{z}_i)$ non-parametrically, we employ a generalized random forest by [Athey et al. \(2019\)](#), adapted to a DiD setting.

We begin by describing the machine learning algorithm used to detect heterogeneity in treatment effects. We then discuss the partitioning variables that condition the cost of displacement.

A. Bird's-Eye View of Machine Learning Algorithm

We now describe the implemented method of a generalized random forest. We seek to estimate the cost of job loss locally at worker and job characteristics $\mathbf{z}$ from equation (2), which is characterized by following local moment conditions:

$$\mathbb{E}(\mathbf{x}'_{it}\epsilon_{it}|\mathbf{z}) = \mathbf{0}_{12}, \quad (3)$$

where $\mathbf{x}'_{it} = [D_i\mathbb{1}_{\{t=t^*+h\}}, \mathbb{1}_{\{t=t^*+h\}}, D_i, 1, \mathbb{1}_{\{t=t^*-5\}}, \dots, \mathbb{1}_{\{t=t^*-2\}}, D_i\mathbb{1}_{\{t=t^*-5\}}, \dots, D_i\mathbb{1}_{\{t=t^*-2\}}]$, ϵ_{it} is the error term from (2), and $\mathbf{0}_{12}$ is a row vector with zeros of length 12. Our approach consists of defining similarity weights $\alpha_{it}(\mathbf{z})$, which measure the relevance of observation it to estimating the cost of job loss at $\mathbf{z}$, and estimating equation

$$\begin{aligned}
(\tau(\mathbf{z}), \theta(\mathbf{z}), \gamma(\mathbf{z})) &= \operatorname{argmin}_{\{\tau, \theta, \gamma\}} \left(\frac{1}{NT} \sum_i^N \sum_t^T \alpha_{it}(\mathbf{z}) \mathbf{x}_{it}' u_{it} \right) \left(\frac{1}{NT} \sum_i^N \sum_t^T \alpha_{it}(\mathbf{z}) \mathbf{x}_{it}' u_{it} \right)', \\
\text{s.t. } \forall_{i,t} u_{it} &= y_{it} - \left(\tau_h(\mathbf{z}_i) D_i \times \mathbb{1}(t = t^* + h) \right. \\
&\quad + \nu_h(\mathbf{z}_i) \mathbb{1}(t = t^* + h) + \theta(\mathbf{z}_i) D_i + \gamma(\mathbf{z}_i) \\
&\quad \left. + \sum_{j=-5}^{-2} \nu_j(\mathbf{z}_i) \mathbb{1}(t = t^* + j) + \sum_{j=-5}^{-2} \delta_j \mathbb{1}(t = t^* + j) \times D_i \right). \tag{4}
\end{aligned}$$

Notice Program (4) takes weights $\alpha_{it}(\mathbf{z})$ as given and is solved for each value of partitioning variables $\mathbf{z}$ separately. For constructing these weights, a generalized random forest is used. For exposition purposes, the algorithm is presented in three steps. First, we show how a single tree in the spirit of Breiman et al. (1984) with a modified splitting criterion borrowed from Athey et al. (2019) is grown. Next, the approach is augmented to generalized random forests. Finally, we present the detailed numerical implementation and how the weights can be recovered from our random forest.

A.1. Tree Construction

The tree-based procedure partitions the dataset into subsamples with similar within-group earnings losses while maximizing between-group differences. Due to the computational complexity, a top-down, greedy approach is traditionally used. The tree-building procedure follows Algorithm (1) recursively. In each partition (called as a node or a leaf), the treatment effect is estimated from Equation (2).⁹

The main difference of our procedure from the textbook CART algorithm Breiman et al. (1984), is the splitting criterion. In the original approach, the algorithm aims at building a tree minimizing the squared sum of residuals.¹⁰ In our setup, we are interested in growing a tree that explores the underlying heterogeneity of earnings losses between partitions of individuals with different characteristics. For this reason, we adapt the criterion (5) proposed by Athey et al. (2019). This criterion maximizes between-group differences of earnings losses, $(\tau_l - \tau_r)^2$, with an adjustment for more balanced splits, $\frac{n_l \cdot n_r}{N^2}$.

⁹While all parameters from (2) are estimated, only the scarring effect enters the splitting criterion (5).

¹⁰In regression trees, the squared sum of residuals is defined as $\sum_j \sum_{i \in \mathcal{D}_j} (y_i - \bar{y}_{\mathcal{D}_j})^2$, where $\mathcal{D}_j$ is a data subset obtained through sample partitioning procedure and $\bar{y}_{\mathcal{D}_j} = \frac{1}{|\mathcal{D}_j|} \sum_{i \in \mathcal{D}_j} y_i$ is the mean of the response variable in the specific set of data.

Algorithm 1 Tree Algorithm of Recursive Partitioning

- i. Start with the whole dataset and consider it as one large data partition, $\mathcal{P}$.
- ii. For each partitioning variable z_k and its every occurring value $\bar{z}$, split partition $\mathcal{P}$ into two complementary sets of individuals i such that $\mathcal{P}_l = \{i \in \mathcal{P} : z_{ki} \leq \bar{z}\}$ and $\mathcal{P}_r = \mathcal{P} \setminus \mathcal{P}_l$ and estimate cumulative earnings losses τ_l and τ_r for both partitions by running two separate regressions of form (2) on $\mathcal{P}_l$ and $\mathcal{P}_r$.
- iii. Choose the variable z_k and value $\bar{z}$ that maximizes:

$$(\tau_l - \tau_r)^2 \frac{n_l \cdot n_r}{N^2}, \quad (5)$$

where n_l and n_r are sizes of $\mathcal{P}_l$ and $\mathcal{P}_r$ and N is the sample size of $\mathcal{P}$.

- iv. If (5) is smaller than a tolerance improvement threshold, then stop. Otherwise, go to step (ii) and repeat the splitting procedure for $\mathcal{P}_l$ and $\mathcal{P}_r$ separately, where $\mathcal{P}_l$ and $\mathcal{P}_r$ are new partitions subject to the splitting procedure, $\mathcal{P}$.
-

An alternative approach assumes the data-generating process is given and estimates accordingly. This would require arbitrarily specifying the functional form of $\tau(\mathbf{z}_i)$. Instead, we explicitly acknowledge our agnosticism about $\tau(\mathbf{z}_i)$ and our procedure simultaneously: (i) identifies which variables contribute to heterogeneous scarring effects; (ii) detects nonlinearities between τ and $\mathbf{z}_i$; (iii) captures interactions (including higher-order) among partitioning variables.

Partitioning variables are detailed in Subsection B. Here we present the tree algorithm’s output illustrating heterogeneous treatment effects; Section V discusses variable-loss correlations using random forest predictions. Figure 1 shows a decision tree capturing heterogeneity in earnings losses (as proportions of pre-displacement earnings). Each node displays average loss and sample percentage. Starting from the full sample (average loss 36.1%), the tree branches using market share, worker age, manufacturing status, firm wage premia, and firm size. The final tree contains 10 leaf nodes, each defined by 2-5 binary conditions, revealing substantial heterogeneity. Workers displaced from large firms with high market share face losses of 54.4%, while workers under 41 from firms with lowest market share experience losses of only 18.4%. The tree systematically maximizes between-group variation while maintaining sufficient observations per leaf.

A.2. Generalization to Random Forests

Tree-based models offer intuitive graphical illustration but suffer from non-robust estimates and difficult standard error estimation. Random forests (Breiman, 2001) address these limitations by constructing an ensemble of trees through bootstrapping the training data and

randomly selecting features at each split. This produces decorrelated trees that reduce overfitting and improve prediction accuracy, identifying only relationships that consistently appear across bootstrapped samples. Each tree uses an “honest” approach: half of the bootstrapped sample determines partition conditions, while the other half estimates the cost of job loss.¹¹ The forest then generates weights measuring how often individuals fall into the same partitions across trees. These weights solve (4) for each value of $\mathbf{z}$ separately.

Forests’ specification. To analyze heterogeneity in post-displacement outcomes, we study three separate variables: relative earnings losses, log-wage losses, and employment losses. Each outcome is examined from both a short-term (one year after displacement) and long-term (ten years after displacement) perspective, resulting in six distinct dimensions of heterogeneity. For each case, we grow a forest consisting of 10,000 trees, with a minimum leaf size of 800 person-year observations. In each split decision, 7 partitioning variables are randomly selected.

An advantage of random forests over single trees is that standard errors can be computed relatively easily. We follow the algorithm proposed by [Athey et al. \(2019\)](#) and use the *bootstrap of little bags* method introduced by [Sexton and Laake \(2009\)](#). In this approach, trees are randomly assigned to different “bags,” allowing them to be reused multiple times to estimate variability without the need to grow new forests from scratch. This method is significantly more efficient than traditional bootstrap in our context, where growing even a single forest requires substantial computational time.¹² Our bootstrap procedure samples workers; thus, standard errors are clustered at the worker level.

The structure of our random forest allows us to estimate individual cost of job loss conditional on observed characteristics, with a key advantage over standard subgroup analysis: we can examine how losses vary with respect to one variable while holding others constant. In estimating these conditional earnings losses, we focus on variables exhibiting systematic differences, as our forest is regularized to filter out spurious variation driven by noise rather than true effects. While our method identifies variation only along observable characteristics—meaning we cannot detect heterogeneity driven by unobserved factors, which limits our ability to speak to job loss heterogeneity more broadly—the analysis remains highly valuable, particularly for informing policy targeting where eligibility criteria typically depend on

¹¹This ensures we do not document spurious heterogeneity driven by outliers in the splitting process.

¹²For simplicity, we present only the general idea of the bootstrap of little bags here. More formally, standard errors are estimated using influence functions and a variance decomposition that captures variability across groups of trees built on different subsamples. This estimated variance is then combined with a problem-specific curvature term to construct valid confidence intervals. A detailed explanation of the method is provided by [Athey et al. \(2019, pg. 18\)](#), while our implementation is described in Section ?? of the Appendix.

observable characteristics.

A potential concern when combining a generalized random forest with difference-in-differences is that the method requires parallel trends not just to hold on average, but within each forest split. If certain covariates correlate with systematic differences in pre-treatment outcome trends, the algorithm might split on these covariates and generate spurious heterogeneity, misinterpreting pre-existing trajectory differences as heterogeneous treatment effects. To address this concern, we conducted several diagnostic checks.

First, we examined the relationship between pre-treatment trends and estimated losses. Using the year immediately prior to displacement ($t = -1$) as our reference, we correlated the group-time fixed effect for the treatment group five years before displacement (δ_{-5} from Equation 2) with estimated post-treatment losses (τ_h) for $h \in \{1, 10\}$. A strong negative correlation between δ_{-5} and τ_h would suggest that estimated losses might be driven by pre-existing trends rather than causal effects. Figure ?? in Appendix ?? displays the relationships, showing no systematic correlation for earnings, employment, wage losses. If anything, we find a *positive* correlation between pretrend coefficients and losses for small intervals of long-term employment and short-term wage losses. This correlation has the opposite sign of what we would expect if pretrends were driving our heterogeneity results, suggesting that any mild violation would bias our estimates toward understating heterogeneity rather than overstating it. The following additional checks put these into perspective. Using the single regression tree presented in Figure 1, we plotted event study coefficients separately for each final node, revealing flat pre-treatment trends across all nodes (Appendix ??). This confirms that even after numerous splits, the parallel trends assumption cannot be rejected based on pre-trends. Finally, we grouped observations by predicted loss deciles and estimated separate event studies (Figure ?? in Appendix ??). In general, the pretrends are very small, and we find no systematic association between pre-treatment fixed effects and estimated loss magnitude. Overall, these findings support our assumption of parallel trends conditional on partitioning variables, and thus our interpretation that the heterogeneity uncovered by the forest approach reflects true variation in treatment effects rather than artifacts from parallel trends violations.

An equally important question concerns the accuracy of the specific loss magnitudes identified through the random forest. Measuring this accuracy presents a methodological challenge, as we are not interested in accurate predictions of an observed dependent variable on the left-hand side in Equation 2 but rather precise identification of an unknown treatment effect, one of the estimated regression coefficients on the right hand side. We address this

concern by proposing an alternative evaluation procedure. In our exercise we create 50 distinctive groups.¹³ For each group, we estimate the average cost of job loss using Equation 1. Then we compare these treatment effects with forest-implied treatment effects. As we show, both measures are very highly correlated, which suggests that our method provides accurate estimates. We relegate a more detailed discussion to Section ?? and Figures ?? and ?? in the Appendix.

B. Partitioning Variables

Before we present our estimation results in the next section, we discuss and motivate our choices for partitioning variables below. What are the reasons that earnings losses differ across individuals? On the one hand, workers in different environments face different re-employment probabilities after job loss, on the other hand, displaced workers might face different wage declines after re-employment. The workhorse framework in empirical research typically posits that wages consist of multiple components: $\ln(w_{it}) = \alpha_i + \psi_{J(i,t)} + \beta * tenure_{it} + \varepsilon_{it}$. α_i represents the worker fixed effect, $\psi_{J(i,t)}$ the firm specific wage component, $\beta * jobtenure_{it}$ the effect of time spent at the current employer on wages, and ε_{it} represents an idiosyncratic match component.

In on-the-job search models such as Burdett (1978), laid-off workers fall down the job ladder in terms of firm wages and match quality and have to climb back through job-to-job transitions. Thus, earnings losses can vary first, because workers with higher firm wage components or better previous match quality face a steeper fall off the job ladder. In addition, workers with high skills α_i before displacement have more room to face depreciation of these skills. Moreover, workers with higher fixed effects might have accumulated more savings and higher claims to unemployment benefits, which would enable them to be pickier in their job search. This would lead to higher employment losses, but lower wage losses after job loss (Nekoei and Weber, 2017). Motivated by this we estimate the following wage equation (Abowd et al., 1999) on a rolling window of all social security spells in the five years before the event:

$$\ln(w_{it}) = \psi_{J(i,t)} + \alpha_i + \theta_t + x_{it}\beta + \epsilon_{it}, \quad (6)$$

where $\ln(w_{it})$ is the log daily wage of the dominant employer at period t , $\psi_{J(i,t)}$ represents the establishment fixed effect of the employer of worker i at period t , α_i the worker fixed

¹³Our groups are created based on the forest-based treatment effects as we find it the most natural. That said, other grouping criteria could be potentially used as well.

effect, θ_t the year fixed effect, and x_{it} is a cubic polynomial of age. We then construct deciles of the estimated firm fixed effect $\hat{\psi}_{J(i,t)}$ and worker fixed effect $\hat{\alpha}_i$, and the wage residual $\hat{\varepsilon}_{it}$, which are then used as partitioning variables.

We further include job tenure and the firm specific separation rate in the five years leading up to the mass layoff event as partitioning variables.

In addition, earnings losses might vary because workers face different job finding probabilities or job offer distributions. These might change for example with the business cycle. To capture all time-varying factors we include a dummy for each displacement year as partitioning variables. Further we proxy the average job offer distribution displaced workers face by computing the average percentile of the firm wage effect in the local labor market.¹⁴ In addition we study how the average unemployment rate over our study period in the local labor market correlate with earnings losses.¹⁵ Workers in concentrated labor markets or separating from a monopsonist employer might face more difficulty to quickly find a similarly good job (Gathmann et al., 2020). Thus, we include the overall local labor market concentration as measured by the Herfindahl-Hirschman index, the firm size, and the labor market share of the previous employer as partitioning variables. We also include a dummy variable for each NACE-1 industry to capture any time-invariant differences in the cost of job loss across industries.

Workers with higher job mobility in the past might find it easier to climb back the job ladder. The number of previously held jobs is therefore included as a partitioning variable, as well as a number of socio-demographic factors such as worker age, a dummy for Austrian citizenship, gender, a dummy for working in a blue collar occupation, and firm age.

To enhance interpretability, we categorize all continuous variables into deciles according to the overall distribution of Austrian workers on our reference day, and not only the selected displaced worker sample. This way the heterogeneity is easily interpretable in terms of the overall employment distribution in Austria. Table ?? in the appendix summarizes all the definitions of the partitioning variables, and Figure ?? reports a correlogram of all partitioning variables.

¹⁴Specifically, we compute the average firm wage premia of all jobs in a given region leaving out all jobs of the worker's current employer. Formally for every worker i employed at firm $J(i, t)$ we compute $\sum_{k \notin J(i, t) \wedge k \in r(i)} \hat{\psi}_{J(k, t)} / \#(k \notin J(i, t) \wedge k \in r(i))$, where $r(i)$ is the region of the worker i .

¹⁵We use the NUTS-3 district (35 categories) for the local labor market

V. HETEROGENEOUS COST OF JOB LOSS

In this section we use the machine learning algorithm to document the unequal consequences of job loss along three outcome variables: losses in relative earnings, employment days, and log wages in the short-term ($h = 1$) and the long-term ($h = 10$).

A. The Distribution of the Cost of Job Loss

The generalized random forest provides a CATE for each individual based on their characteristics. We start by plotting the distributions of τ_h for relative earnings, employment, and wage changes, for horizon $h \in \{1, 10\}$. Figure 2 shows the histogram and boxplots of the CATEs for the three variables for the short-run (left panels) and long-run (right panels). The figure clearly shows that the short and long-term consequences of job losses are far from uniform across individuals. While the median worker in our sample is facing predicted earnings losses of 35 percent one year after the job loss, some workers lose less than 20%, while the hardest hit are predicted to lose over 70%. The 25th percentile of CATEs is 42%, while the 75th percentile is 29%. The distribution of long-run losses shifts towards lower losses, and its dispersion shrinks markedly. While few workers face predicted long-term losses of more than 25 percent, the interquartile range narrows from close to 15pp in the short run to 6pp in the long-run.

This heterogeneity is not driven by noisy estimates of the CATEs. At the 95% confidence level, we estimate that 67% of workers face significantly different predicted short-run earnings losses compared to the median of 35%. Because long-run losses are less dispersed and also somewhat less precisely estimated, close to 40% of long-run CATEs are significantly different compared to the median of 15%.

Earnings losses are the result of both forgone earnings during non-employment and declines in wages after reemployment. In the second and third row of Figure 2, we plot the distributions of CATEs of employment and log-wage losses in the short and long term. CATEs of employment losses one year after job displacement range from more than 150 days for the 90th percentile to 73 days for the 10th percentile. Similar to overall earnings losses, the dispersion of employment losses is significantly lower 10 years after the mass layoff event; employment losses are concentrated between 50 and 16 days per year, 80 percent of workers are predicted to suffer long-term employment losses in this range.

The last row shows that CATEs for wage losses in the short run range from 35% to even wage gains for a handful of individuals. Interestingly, the distribution of CATE wage losses

in the short and long run are very similar, suggesting very persistent wage losses.

This is also illustrated quantitatively in Figure 3a, where we present the correlation matrix between short- and long-run CATEs of earnings, employment, and wage losses. Individuals with the highest predicted cost of job loss in the short run also face higher losses in the long run. CATEs for earnings and employment losses one and ten years after the job loss have a correlation of between 0.40 and 0.45. Predicted wage losses are much more persistent; short- and long-run CATEs are highly correlated with 0.75. The correlation pattern also reveals whether employment or wage losses are the main drivers of overall earnings losses. In the short run, earnings losses are more correlated with employment losses (0.94) than with wage losses (0.52). Although the correlation of wage and earnings losses increases in the long run somewhat, long-run earnings losses are still more related to employment than to wage losses, with the correlation coefficient being 0.85 versus 0.55.

Remarkably, even 10 years after the initial job displacement almost all workers are predicted to experience earnings, employment, and even wage losses conditional on re-employment. One would perhaps expect that workers previously employed in low-wage jobs with high turnover would improve their outcomes relative to the control group, but there is essentially no combination of our partitioning variables that predict this to happen. Thus, in line with recent evidence from the United States (Rose and Shem-Tov, 2023), job loss is costly even for low wage workers with disadvantaged worker and job characteristics.

A long-standing concern in the job displacement literature is that the strict sample selection criteria typically applied select workers that are bound to experience high earnings losses. This raises concerns about the generalizability of the documented earnings losses to the overall population. To tackle this, we use our grown forests to predict earnings losses for each employed worker that did not satisfy our sample criteria given their worker and job characteristics.¹⁶ Figure 3b compares the distribution of long-term CATEs for earnings losses in our displaced worker sample and the general population. The overall distributions are very similar. The differences in the 10th, 25th, 50th, 75th, and 90th percentiles of CATEs are all within 1pp. At least in our sample, the sample restrictions do not select worker and job characteristics that are related to significantly higher long-run earnings losses.

¹⁶For workers who have lower job tenure or are employed at smaller firms than our sample restrictions, we assume the lowest possible values.

B. Post-Displacement Evolution

Why do workers face such different consequences from mass layoffs? Before we turn to which partitioning variables are the most important predictors of the cost of job loss in the next section, we study how labor market trajectories after the job loss differ for low versus high predicted earnings loss individuals. We bin workers into quartiles according to their short-run CATE in earnings losses and estimate the full event-study specification of Equation (1) for various left-hand side variables separately for every CATE quartile.

Figure 4 plots the dynamic evolution of earnings losses, employment losses, log-wage losses, and firm wage premia losses over time. The figure reconfirms that displaced workers experience vastly different labor market outcomes after job losses and that the heterogeneity in the cost of job loss is persistent. Although we group workers by their predicted earnings losses in $t + 1$, only the quartile with the highest losses (Q1) experiences any catch-up in earnings and employment. Wage losses are more persistent; there is barely any convergence noticeable between the groups.

Panel (c) and Panel (d) show the cost of job loss in terms of log-wages and firm wage premia measured in log points. Recently, a number of papers have studied the contribution of losses in firm wage premia to wage losses.¹⁷ Changes in the firm wage component explain between 60-70 percent of the overall change in wage losses for each group. Changes in the firm wage premia also explain about 60-70 percent of the between-group differences in the evolution of wages after job loss. Thus, in our setting, the contribution of firm wage losses is rather uniform over the earnings-loss distribution.

Panel (a) of Figure 5 shows that differences in match quality, as measured by the residual in the AKM Mincer wage regression Equation (6), cannot explain the large wage loss differences.

One explanation for wage losses is that workers lose firm- and industry-specific human capital.¹⁸ Indeed, Panel (b) shows that the higher the predicted losses, the less likely it is that workers stay in their original NACE-1 industry. Any job- and industry-specific human capital losses might be compounded if workers in addition cannot find stable employers after job losses (Jarosch, 2023). Panels (c) and (d) show that workers with the highest short-run CATEs in earnings losses move to firms with less job security and have more job changes consequently.

¹⁷*E.g.*, Fackler et al. (2021), Schmieder et al. (2023), Bertheau et al. (2023), and Lachowska et al. (2020).

¹⁸See *e.g.*, Jacobson et al. (1993), Neal (1995), and Jarosch (2023),

C. Correlates of Earnings Losses

What are the most important pre-displacement worker and job characteristics that predict the cost of job loss? These variables could be used to target policy interventions, something we tackle in the subsequently.

An intuitive way to judge the importance of individual factors is to compute the elasticity of losses with respect to individual variables, holding all other factors constant. We compute how CATEs in earnings, employment, and wage losses change with different values of one variable at a time, while holding all other variables fixed at their median. This way we are able to control for observable confounding factors. Figure 6 provides an overview by how much CATEs change, when moving individual partitioning variables from the lowest (min) to its highest (max) value, while holding other variables constant at their median. We present the point estimates of our forests, alongside the 95 percent confidence intervals that take into account both the uncertainty arising from the estimation and the building of the forests. In appendix section ?? we present the CATEs for each realization of the partitioning variables, but as losses typically exhibit monotone and near-linear relationships with all variables, Figure 6 provides an accurate overall picture. For more readability, we only show the ten most important variables. As can be seen in the figures, the elasticity of predicted losses with respect to less important variables is close to zero.

The first row shows the most important predictor of earnings losses in the short (left panel) and long run (right panel), while the second and third row present the results for employment and log-wage losses. Worker’s age is the most important correlate for short-term losses and the third most important for long-term earnings losses. The oldest workers in our sample face predicted short-term earnings declines of 50 percent, almost twice as high as for the youngest workers, even when we hold all other partitioning variables constant. In the long run the oldest workers face 4pp higher predicted losses.¹⁹ This finding is consistent with Guvenen et al. (2017), who also find that old workers have much higher earnings losses. Do older workers face higher earnings losses because of worse employment or wage losses? Although age is always among the 10 most important variables for employment and wage losses, age plays a much bigger role for employment losses. The oldest workers face predicted employment declines of 165 days in the year following the job loss, while this number is 98 lost employment days for young workers. In the long run a differential of 18 employment

¹⁹Labor force attachment of old workers has risen over our sample period. Did this higher attachment lead to a change in the earnings loss profile over age? To answer this, we predicted the age profile for different displacement years. For short-term losses there is no noticeable change over the years, for long-term losses the age profile flattens somewhat in the 2000s, although this change is very small.

days persists. The differential for wage losses is with 2.5-3.5pp much smaller.

Furthermore, the displacing firms' characteristics play an important role in shaping the cost of job loss. Firm size, the labor market share, the firm's separation rate, and the firm wage premium all are among the most important predictors for earnings, employment, and wage losses.

Firm size is the second most important variable for short-term and market share the most important variable for long-term earnings losses. This is in line with [Fackler et al. \(2021\)](#), who find elevated losses in larger firms in Germany. In our setting, this holds even after controlling for many observable worker and firm characteristics. Workers from the largest firms face earnings declines that are 15pp larger in the short run and 8pp larger in the long run than the losses in the smallest firms.

[Gathmann et al. \(2020\)](#) document that mass layoffs at regional flagship firms can have negative spillover effects on other firms in the region, and thus depress labor market outcomes for the whole region. Our results confirm this: separations from firms with a high labor market share in a given region depresses long-run earnings, mostly through more persistent employment losses. Workers separating from firms with the highest employment share face close to 20 more lost employment days 10 years after the job loss.

Recently, the role of firm wage premia for the cost of job loss moved to the center of attention in the literature. [Fackler et al. \(2021\)](#), [Schmieder et al. \(2023\)](#), and [Bertheau et al. \(2023\)](#) find losses in firm wage premia explaining a large fraction of wage losses, whereas [Lachowska et al. \(2020\)](#) find limited role of firm wage premia. We shed light on this discussion from a new angle: Is the pre-displacement firm wage premium an important predictor for subsequent earnings, employment, and wage losses? For wage losses, the answer is clearly yes. The firm fixed effect is the most important variable for short and long-run wage losses. Workers being displaced from the highest paying decile of firms face short-term losses of 18pp and long-run losses of 9pp more than the employees of the lowest paying firms. Thus, for each log point higher pre-displacement firm fixed effects, long-run wage losses increase by roughly 0.15 log-points. Although the firm fixed effect is not associated with higher employment losses, due to its strong correlation with wage losses, it features as the third and fifth most important variable for earning losses, with loss differentials of 10pp and 3pp for the short and long run, respectively.

Consistent with the theory in [Jarosch \(2023\)](#), workers who lose jobs at particularly stable firms are predicted to face 8pp higher earnings losses. This is the second highest elasticity for long-run earnings losses. These higher earnings losses are a combination of higher em-

ployment (12 days) and wage losses (3.4pp).

In contrast to employment losses, wage losses vary strongly with the pre-displacement person fixed effects and gender. When comparing workers with the same characteristics, females are predicted to face 6.5pp higher short and 2.5pp higher long-run wage losses than men, similar to findings in Illing et al. (2024). The person fixed effect is even more important. Workers in the highest decile of worker fixed effects are predicted to experience 11pp higher wage losses in the short run and 3pp higher wage losses in the long run compared to the lowest.

Few of the remaining variables show notable associations with losses. This is surprising, as some of these variables are discussed prominently by the prior literature. One example is business cycles, Schmieder et al. (2023), Davis and Von Wachter (2011) among others point to large differences in losses over the business cycles. Over our study period, Austria faced several recessions: 1995, 2003, 2009 all featured a decline in GDP together with rising unemployment rates. Despite this, year dummies do not emerge as important predictors for losses. This suggests that in our setting, changes in displaced workers' characteristics drive outcomes more than the recession itself.²⁰

The literature has documented elevated losses of high-tenured workers and has taken this as evidence for the important role of human capital depreciation for explaining losses (Jacobson et al., 1993; Neal, 1995). We in contrast do not find job tenure to be an important predictor for the cost of job loss.

Helm et al. (2023) document that manufacturing workers face larger than average earnings losses. In our setting, once we control for different firm and worker characteristics of displaced workers across manufacturing and non-manufacturing workers, manufacturing is no longer special. Generally speaking, none of the industry dummies with the exception of construction shows up among the most important correlates.

Perhaps surprisingly, regional characteristics such the regional unemployment rate, the Herfindahl index of labor market concentration, and average firm pay premia in the region do not feature often among the most important variables. The only exception is the Herfindahl index, very concentrated labor markets are associated with 15 days of additional long-run employment losses. Summarizing, losses are more strongly correlated with firm and worker characteristics than with environmental factors.

One potential criticism of the partial effects presented so far is that an individual with

²⁰We study this in more detail in an accompanying paper for the great recession and the Covid recession (Gulyas and Pytka, 2020).

median characteristics might not be representative for the whole population, and presented effects might be very different for characteristics other than the median. To tackle this critique, we use partial dependence plots proposed by Friedman (2001) and estimate the earnings losses for each individual by changing the value of one variable, while holding all other characteristics constant at their empirical values, and then averaging over individuals. The appendix shows that this yields very similar results.

An alternative and popular way in the machine learning literature to assess which factors are the most important is the occurrence frequency of variables in the splitting criteria. Figure ?? in the appendix shows that the overall picture for the most important predictors is very similar using this method.

VI. POLICY TARGETING

Given substantial heterogeneity in displacement costs across observable characteristics, labor market interventions may be more effective when targeting workers predicted to face high losses. While our forests could directly identify such individuals, many forest variables are not easily interpretable by policymakers or the public. We therefore construct classification trees using only straightforward criteria (age, pre-displacement income, firm size) while excluding complex variables (firm wage premia, regional concentration metrics, match effects) and year dummies. We project the losses identified by our random forest onto these simpler classification tree structures from Breiman et al. (1984), creating transparent, time-invariant decision rules. This approach ensures “the right to explanation” for those affected by policies while enabling effective targeting of interventions to workers facing employment or wage losses.²¹

Figure 7 presents classification trees that identify individuals with above median long-term losses. These correspond to workers with at least a 10% drop in wages and employment losses of at least 28 days. The red color indicates leaves where at least 50% of the workers experience above-median losses. As shown in Panel 7a, labor market program especially effective at mitigating long non-employment spells should target the following groups: (i) employees older than 38 displaced from firms with above-median market share, (ii) workers

²¹In EU countries, policymakers must provide grounds for their decisions. The General Data Protection Regulation (EU) 2016/679 introduced by the European Parliament and Council of European Union (2016) established “the right to explanation” for individuals subject to automated decision-making. Article 12 requires “*concise, transparent, intelligible and easily accessible form, using clear and plain language*” for information processing, while Article 13 guarantees the right to “*meaningful information about the logic involved.*”

younger than 38 displaced from firms in the top decile of market share or who had relatively few previous employers before displacement. Using these relatively simple targeting criteria, policymakers could correctly identify 75.6% of individuals who will experience above-median employment losses.

Regarding policies addressing severe long-term wage losses, Panel 7b indicates that resources should primarily target the following groups: (i) workers with pre-displacement income in the top quintile who are either older than 38 or displaced from large firms, (ii) workers outside the top income quintile who are older than 37 and are blue-collar or displaced from large firms. This targeting approach would correctly identify 77.3% of workers who will face above-median wage losses. The exercise in this section illustrates a potential application for guiding optimal policies. In principle, alternative trees can be easily built using different loss thresholds and variables.

VII. HETEROGENEITY DETECTION: COMPARISON WITH OTHER METHODS

In this section, we compare the short-term losses identified through our generalized random forest with alternative methods for studying treatment effect heterogeneity. We adopt two complementary perspectives. First, we examine heterogeneity related to observed covariates by estimating a difference-in-differences regression with LASSO-selected covariate interactions, an increasingly popular approach. Second, we consider overall heterogeneity without restricting to observables, employing two quantile-based DiD estimators—Changes-in-Changes and quantile DiD from [Athey and Imbens \(2006\)](#), which allow us to identify the full distribution of treatment effects. We also use the statistical “twins” method from [Schmieder et al. \(2023\)](#), which matches treated and control individuals using propensity scores.

One strategy for studying heterogeneity in earnings losses is to interact all coefficients from Equation (1) with observable characteristics. Fully interacting all partitioning variables would require estimating 64,795 parameters, risking overfitting. To address this, we implement LASSO ([Tibshirani, 1996](#)) penalization on all interaction terms.²² This regularization reduces the number of parameters to 5,402.

Table 3 shows the distribution of identified losses and their accuracy. The interquartile range for LASSO treatment effects is approximately 30pp, twice as large as the 15pp observed

²²Details are presented in Appendix ??.

with our generalized random forest. However, LASSO estimates are less precise: only 21% of individuals have losses statistically different from the median, compared to 67% with the random forest approach. Moreover, the LASSO median treatment effect (-0.100) differs substantially from the homogeneous effect (-0.361) estimated previously.

One explanation for these differences is that LASSO’s objective function prioritizes predicting the *dependent* variable rather than treatment effects. Since earnings losses are small relative to earnings levels, the algorithm may emphasize variables explaining levels rather than treatment heterogeneity.

Another approach for detecting heterogeneity comes from the statistical twins method of Schmieder et al. (2023). This method pairs each displaced worker with a matched control who retained employment, then calculates individual-level DiD estimates by comparing short-term changes between the two. As shown in Table 3, this identifies a much wider distribution of losses, with an interquartile range of 80pp, substantially higher than GRF or LASSO. However, the approach has important limitations: it struggles to filter statistical noise, provides no standard errors, and unlike GRF, cannot predict losses from covariates.

The final approach uses quantile regression to detect variation in earnings losses independently of observed characteristics. We employ two non-linear DiD extensions, Changes-in-Changes (CiC) and Quantile Difference-in-Difference (QDiD), which identify changes in the outcome distribution caused by treatment.²³ The treatment effect at quantile q is the difference between the q -quantiles of treated and control groups. This method is popular for studying distributional changes of earnings, as exemplified by Jarosch (2023, Section 4.1.4).

The left panel of Figure 8 shows that the distribution of individual earnings losses from our forest is much less dispersed than changes in distributions identified through CiC and QDiD. While CiC and QDiD effectively evaluate overall distributional responses to treatment, they impose stringent requirements for identifying individual-level effects. Quantile regressions can only accurately assign individual earnings losses if the ranks of earnings before and after treatment remain perfectly correlated — a condition rarely met in practice. The right panel of Figure 8 clearly illustrates this limitation in our data and strong mean reversion.²⁴ Workers’ pre-treatment ranking of the dependent variable (relative income), on average falls down after job separation, while the pre-treatment bottom 50% climb up in their ranking.

²³A detailed discussion on how to impute the comparison group in the DiD setting can be found in Athey and Imbens (2006), where CiC and QDiD were first proposed.

²⁴This violation of rank preservation leads to non-monotone estimates for quantile effects, as shown in the left panel of Figure 8. The effect occurs because the difference between the 20th quantile of the treatment group and the control group is higher than the differences observed at higher quantiles.

Consequently, these methods capture distributional changes, but they do not reveal the distribution of individual losses.

VIII. CONCLUSIONS

We implement a generalized random forest (Athey et al., 2019) to a DiD setting to study the heterogeneity of earnings losses of displaced workers across observable characteristics. This methodology allows us to make a number of important empirical contributions to the existing literature.

First, we document substantial predictable heterogeneity in the cost of displacement across individuals. Almost no worker is predicted to experience declines lower than 20% in the year after displacement, while the worst affected workers face losses exceeding 70%. Although losses converge across workers over time, short and long-term losses remain correlated at 0.4 at the individual level. Essentially all workers face long-run earnings declines, suggesting displacement is costly even for low-paid workers with disadvantaged characteristics. Heterogeneity in earnings losses is driven primarily by employment rather than wage reductions, though wage losses themselves are highly persistent (correlation of 0.8 between short and long-run).

Second, a long-standing concern is whether the strict sample restrictions used in the literature select individuals that are bound to experience large earnings losses. We use the random forest to predict long-term earnings losses for the population that fails to meet the sample selection criteria and show that they are surprisingly similar.

Third, we show that the most important predictors for overall earnings losses are worker's age, and firm characteristics such as size, employment share in the region, its job security before the mass layoff event, and its wage premia.

We further derive easily understandable decision rules to target specific programs to workers predicted to face long non-employment spells and long-run declines in wages. We cannot speak to the welfare effects of active labor market programs, but our results suggest that policies aimed to mitigate the consequences of job loss such as firm bailouts, employment protection, and employment subsidies such as short time work schemes, which are often applied universally, should be targeted.

IX. DATA AVAILABILITY

Code replicating the tables and figures in this article can be found in Gulyas and Pytka (2026). We use confidential Austrian social security records from 1984 through 2020, which comprise day-to-day information on all jobs and unemployment spells covered by social security. This data (Arbeitsmarktdatenbank, 2021) can be obtained for scientific use, see the replication files for more detail.

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	Displaced	Selected Control Group	Not Selected
Age	38.4	38.32	38.46
$\log(w_{t-1})$	4.49	4.5	4.61
$\log(w_{t-2})$	4.47	4.48	4.59
$\log(w_{t-3})$	4.45	4.46	4.56
Job Tenure (in days)	2770.23	2749.75	2916.08
Firm Size	222.4	213.64	825.47
Manufacturing	0.56	0.56	0.46
Female	0.41	0.41	0.37
Whole Yr. Non-Emp.	0.02	0.02	0.02
Days Non-Emp.	44.36	43.51	39.06
No SS Spell	0.01	0.01	0.02
Person FE (Perc.)	47.79	51.38	57.74
Obs	50207	50207	11704372

Table 1: Sample characteristics of displaced workers compared to the selected control group via propensity score matching and the universe of worker/year observations satisfying sample restrictions

Table 2: DiD Regression

	Earnings		Employment		Log wage	
	(1)	(2)	(3)	(4)	(5)	(6)
τ	-0.361 (0.002)	-0.361 (0.002)	-106.054 (0.743)	-106.054 (0.743)	-0.140 (0.002)	-0.136 (0.002)
Worker FE	No	Yes	No	Yes	No	Yes
Num. obs.	602484	602484	602484	602484	587727	587727
R ²	0.212	0.385	0.235	0.396	0.012	0.917
Adj. R ²	0.212	0.262	0.235	0.275	0.012	0.900

Table shows estimation results using equation (1) for horizon h=1 with using different controls. Earnings refer to earnings/pre-treatment earnings. See text for details.

	Treatment Percentiles			Fraction of individuals with losses <i>statistically significant</i> :		
	25%	50%	75%	below the median	above the median	below or above the median
GRF	-0.42	-0.35	-0.29	0.34	0.33	0.67
LASSO	-0.26	-0.10	0.05	0.11	0.10	0.21
Statistical “twins”	-0.81	-0.31	-0.01		–	

Table 3: Heterogeneity in earnings losses and estimation precision. The left panel shows the 25th, 50th, and 75th percentile of earnings losses for LASSO, generalized random forest, and statistical “twins” methods. The right panel presents the fraction of workers with statistically significant losses different from the median at the 95% confidence level. The median treatment effect used to determine statistical significance is calculated separately for each method. Note that the statistical “twins” approach, borrowed from [Schmieder et al. \(2023\)](#), does not provide a straightforward way to estimate standard errors for individual effects, hence the missing values in the right panel.

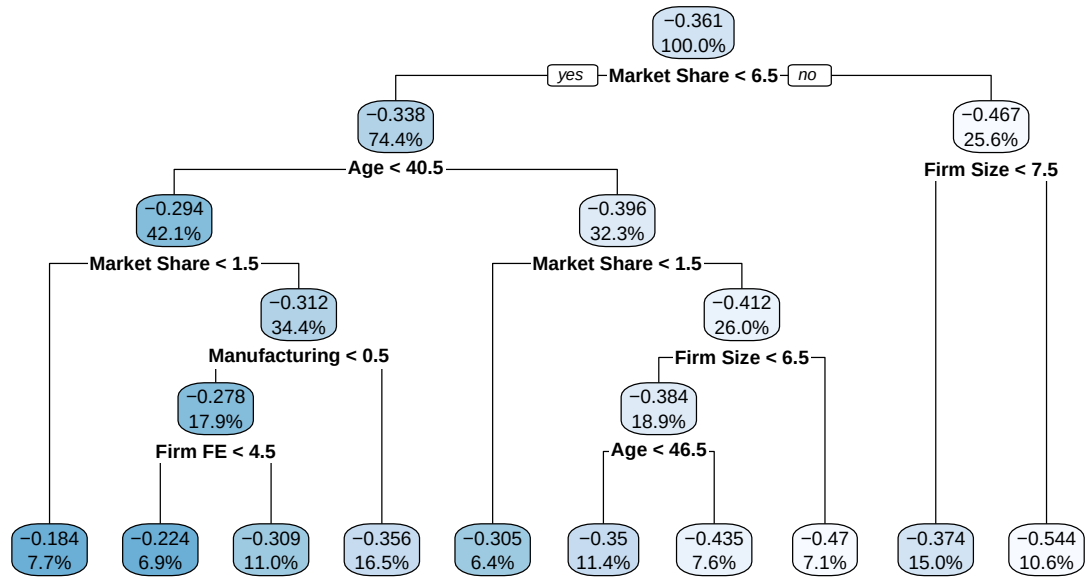


Figure 1: **Heterogeneous earnings losses from job displacement.** Decision tree showing earnings losses as proportions of pre-displacement earnings. The tree partitions workers into 10 groups using market share, worker age, manufacturing status, firm wage premia, and firm size. Each node displays average earnings loss and sample percentage. Losses range from 18.4% (workers under 41 from firms with lowest market share) to 54.4% (workers from large firms with high market share). Built using CART algorithm maximizing between-group heterogeneity with minimum node size of 35,000 person-year observations.

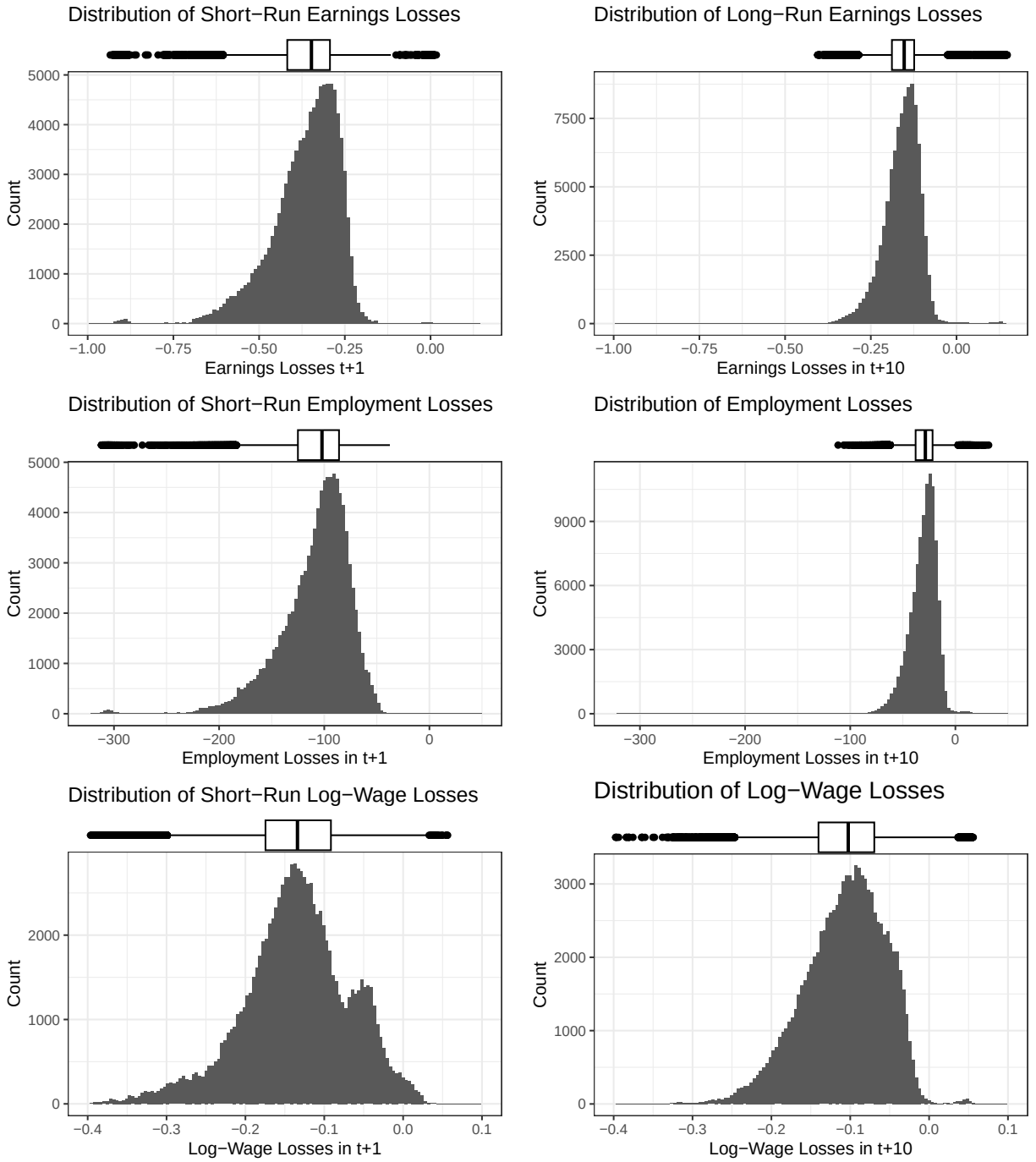
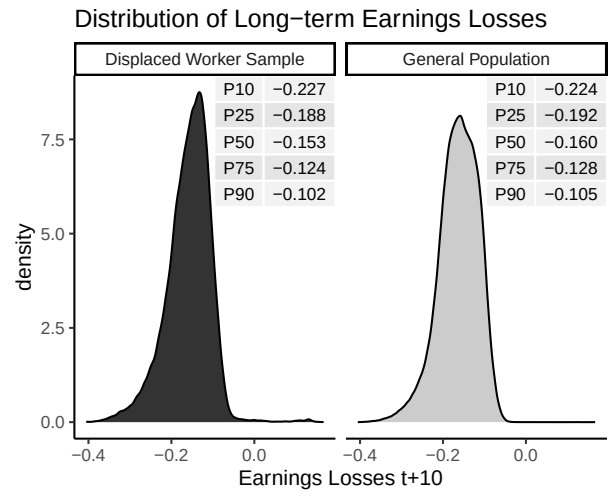


Figure 2: Histogram with accompanying boxplots of short and long-run CATEs of earnings, employment, and log-wage losses. Estimates from a generalized random forest.

	Earnings Loss t+1	Earnings Loss t+10	Empl. Loss t+1	Empl. Loss t+10	Wage Loss t+1	Wage Loss t+10
Earnings Loss t+1	1.00	0.41	0.95	0.40	0.55	0.54
Earnings Loss t+10	0.41	1.00	0.36	0.85	0.33	0.62
Empl. Loss t+1	0.95	0.36	1.00	0.45	0.37	0.41
Empl. Loss t+10	0.40	0.85	0.45	1.00	0.13	0.38
Wage Loss t+1	0.55	0.33	0.37	0.13	1.00	0.76
Wage Loss t+10	0.54	0.62	0.41	0.38	0.76	1.00

(a) Correlations



(b) Distribution of Losses

Figure 3: Left panel: Correlation between CATEs for relative earnings losses, employment losses, and wage losses. Right panel: Distribution of CATE earnings losses for displaced worker sample and the general population. Estimates from generalized random forests.

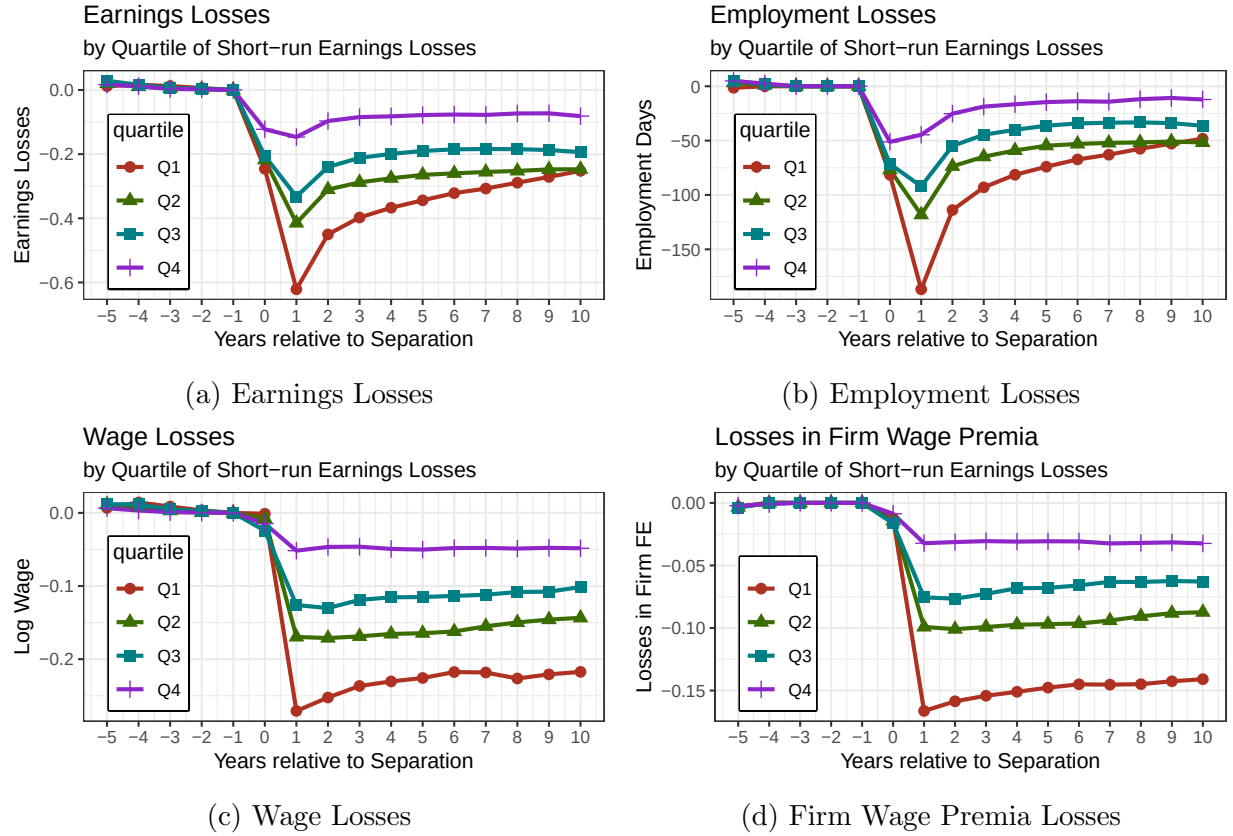
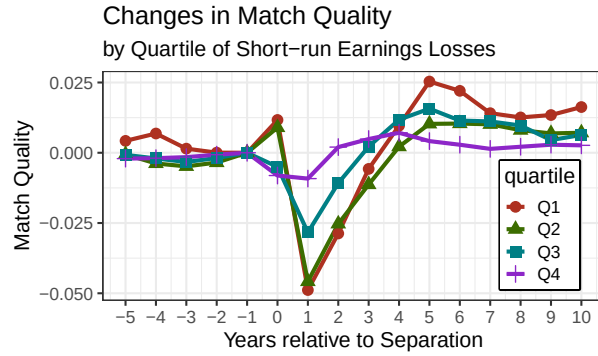
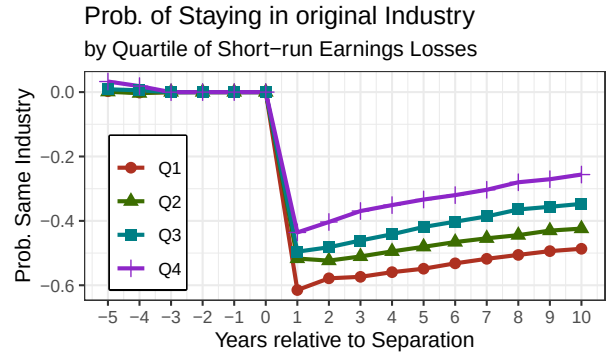


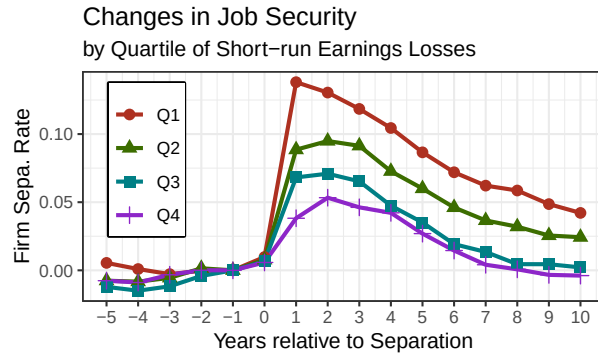
Figure 4: Event-study regression estimates of earnings, employment, wage, and firm wage premia losses, by quartile of predicted short-run earnings losses



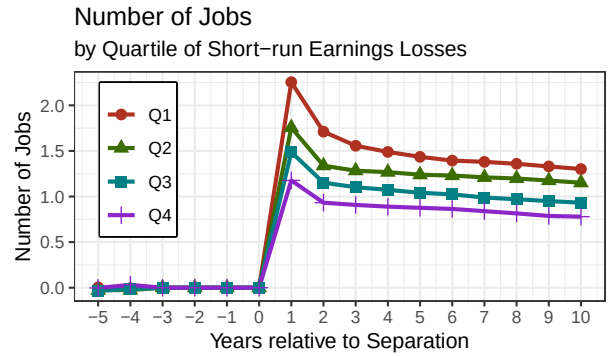
(a) Match Quality Losses



(b) Prob. of Working for Original Industry



(c) Job Security Losses



(d) Number of Jobs

Figure 5: Event-study regression estimates of match quality changes, probability of staying in the same industry, changes in firm separation rates, and number of jobs, by quartile of predicted short-run earnings losses.

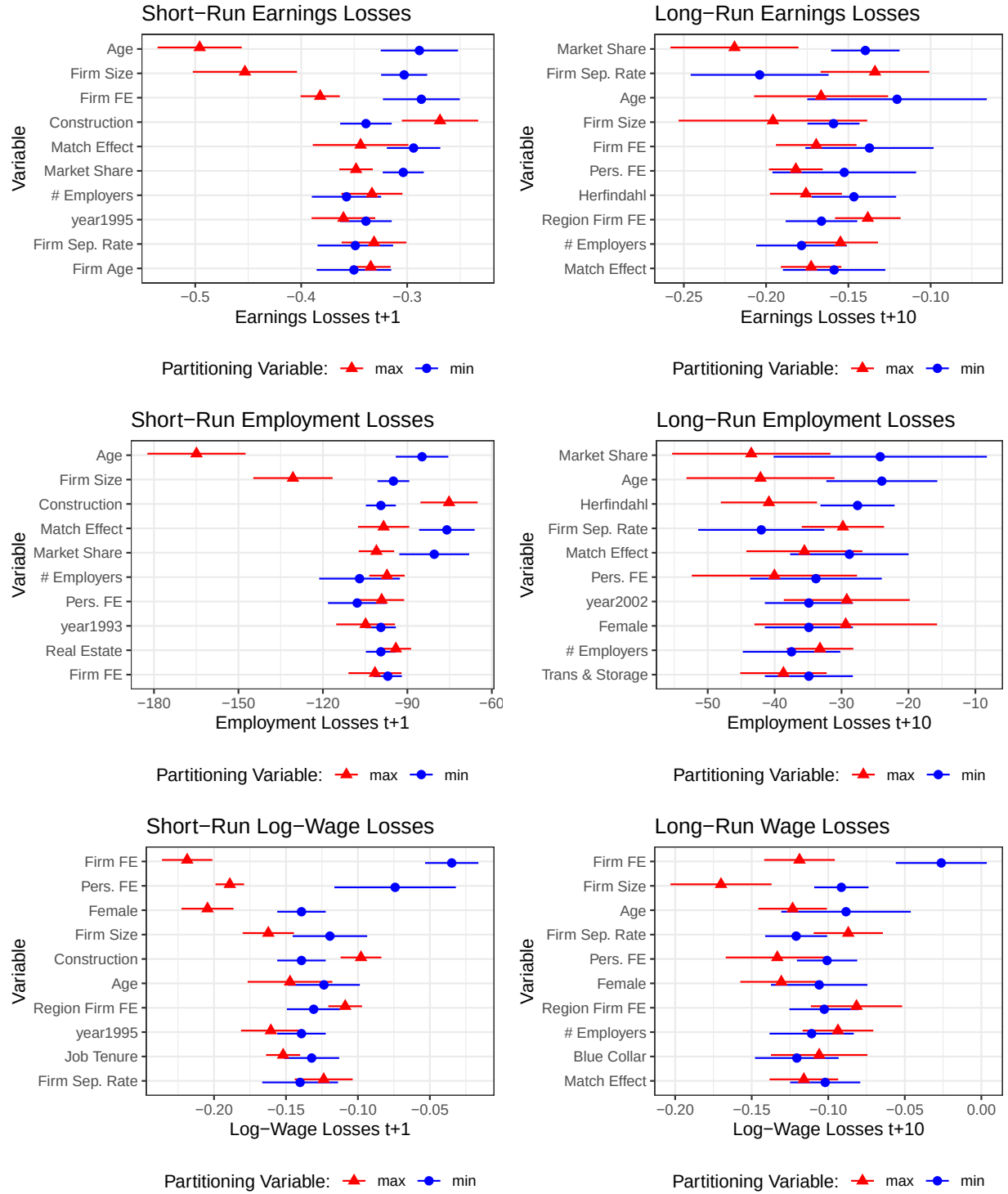
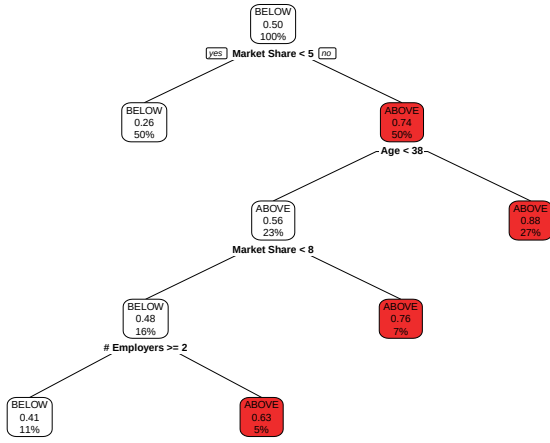
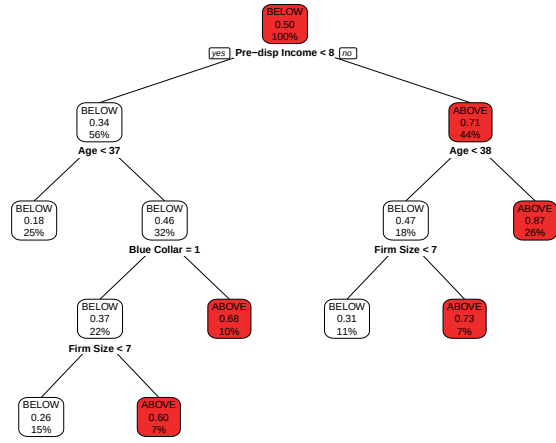


Figure 6: The 10 most important predictors of earnings, employment, and wage losses. Estimates show the Conditional Average Treatment Effect (CATE) from a generalized random forest for the minimum and maximum levels of the partitioning variables, holding all other factors constant at their median values.



(a) Long-term employment losses



(b) Long-term log wage losses

Figure 7: **Policy trees for different long-term losses.** Each leaf in the tree is characterized by three elements: (i) whether individuals in that group are predominantly above or below the median loss (ABOVE or BELOW), (ii) the share of individuals in the leaf who experience above-median losses, and (iii) the share of the total sample represented by that leaf. The root (top) node includes the entire sample, so it always shows 100%. The shares across all terminal nodes sum to 100%. At each split, the left branch groups individuals who satisfy the displayed condition.

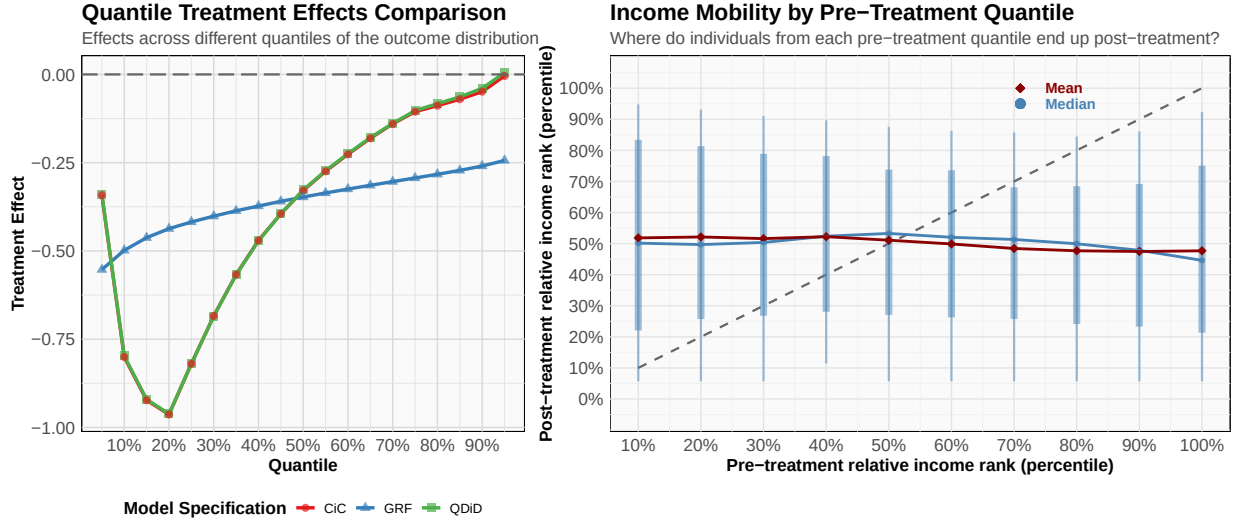


Figure 8: The left panel shows changes in earnings deciles before and after displacement, estimated using QDiD and CiC methods. For GRF, the plot displays deciles of individual treatment effects. The horizontal axis indicates the quantiles of the outcome distribution, and the vertical axis represents the estimated treatment effect. The right panel illustrates the relationship between individuals' pre-treatment relative income rank (5 years before displacement) and their post-treatment relative income rank (one year after displacement). Each vertical line shows the inter-quartile range (25th–75th percentile) and the 10th–90th percentile of post-treatment ranks. The 45-degree dashed line denotes perfect rank persistence.